Double Humanized PD-1&PD-L1 Mouse

Strain Name: C57BL/6-*Pdcd1*$^{em1(hPDCD1)}$ *Cd274*$^{em1(hPD-L1)/Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-00100

Checkpoint blockade is a promising immunotherapy approach to block the function of immune checkpoint proteins. In the recent years, checkpoint blockade therapies, particularly monoclonal antibodies blocking the inhibitory programmed cell death 1 (PD-1/PD-L1) pathway, have achieved significant clinical advances, leading to durable therapeutic responses and long-term remission for a growing number of solid and hematological malignancies.

Construction strategy

On the C57BL/6 background, the full-length coding sequence of human CD274 gene was placed immediately downstream of the start codon of the mouse endogenous Cd274 gene, followed by a poly(A) site. This guarantees an exclusive expression of human PD-L1 in the humanized mice. A similar construction strategy was used for Pdcd-1 gene replacement.

Validation data

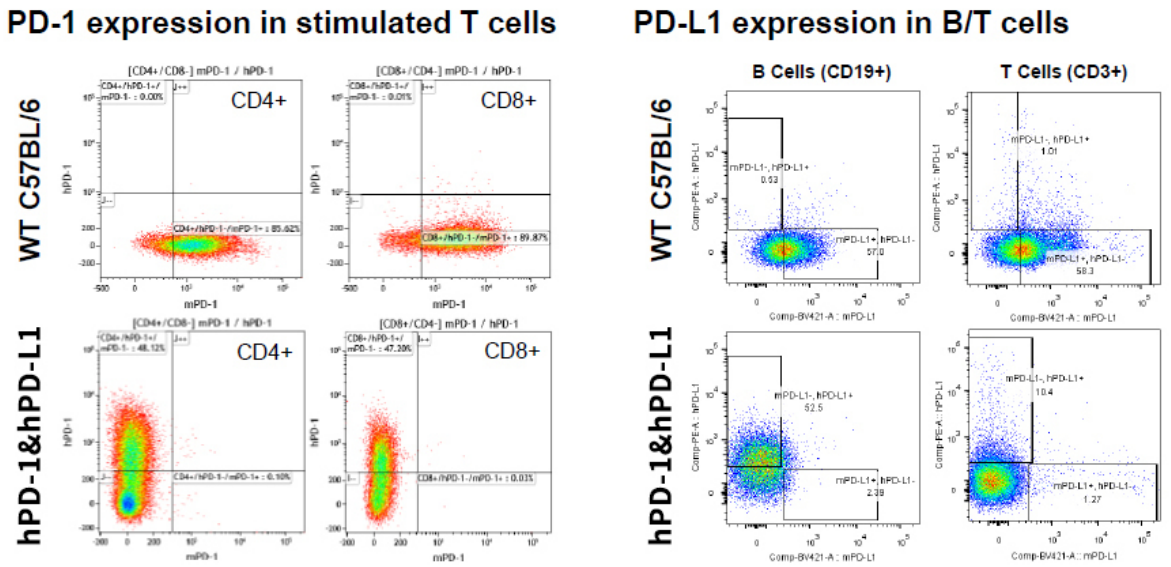

Figure 1. The expression of human PD-1 and PD-L1 in double humanized PD-1&PD-L1 mice was confirmed by FACS.

Efficacy study

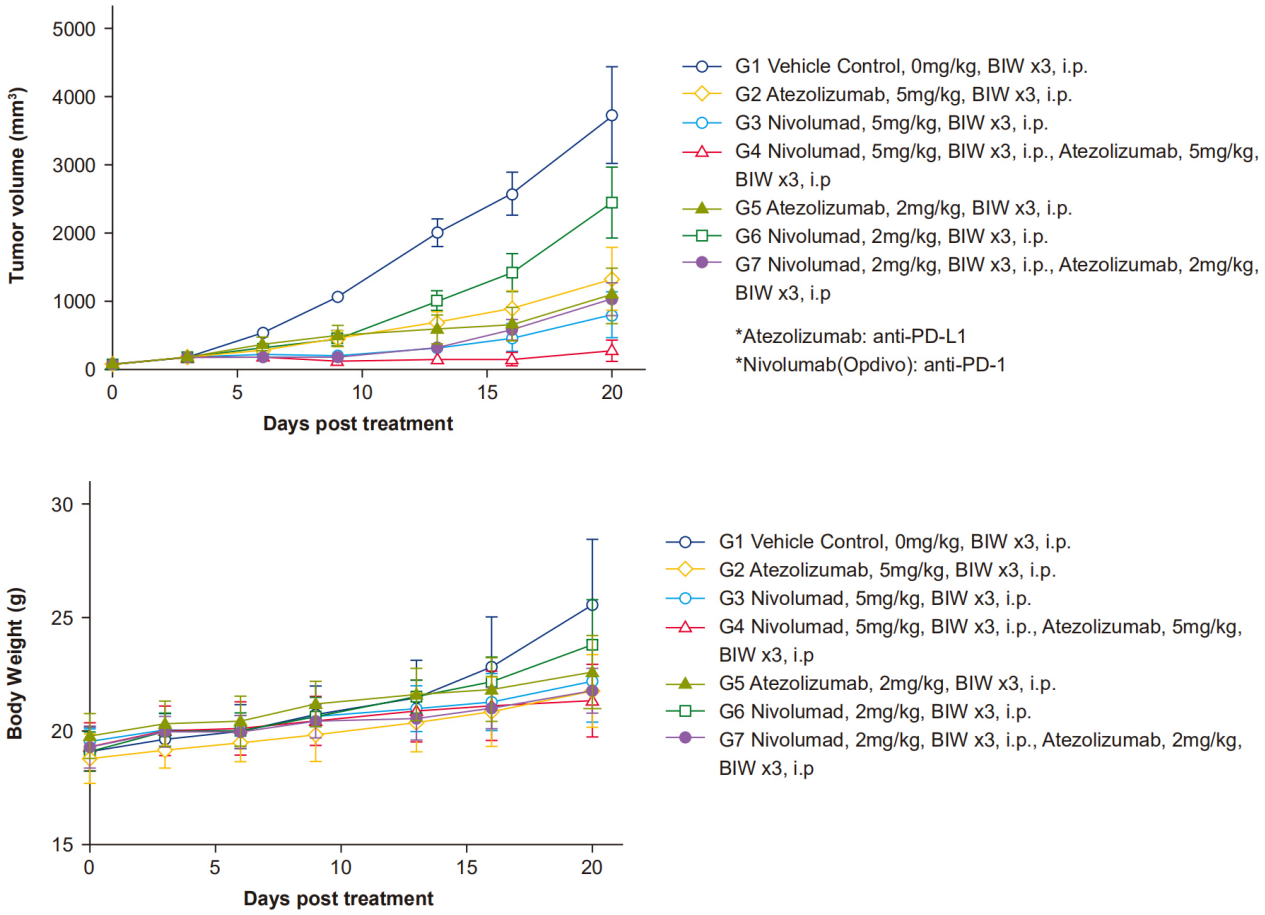

Figure 2. In vivo validation of double humanized PD-1&PD-L1 mice. Double humanized mice were inoculated with MC38 cells, and randomly assigned to different groups (n=8) when the tumor grew to a volume of 100 mm3. A combinatorial treatment of anti-PD-L1 and anti-PD-1 demonstrated a noticeable efficacy improvement compared to the same dose of single agent (top) without affecting the animal body weight (bottom).

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com**
or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

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2200 Smithtown Ave, Ronkonkoma, NY 11779

www.genetargeting.com • inquiry@genetargeting.com